Library Management System: Requirement Prioritization Using the MoSCoW Method

Course code: CSA1006

Course: Software Engineering

Name: B. Pavani

Register No: 192525248

Tool Used: Microsoft Excel

Aim

To categorize the given Library Management System requirements using the MoSCoW method and justify each category based on its impact on users, feasibility, time, budget and resources.

Software Used

Microsoft Excel (the same workbook can also be opened in Google Sheets).

Concept of the MoSCoW Method

MoSCoW is a method used to decide which requirements should be delivered first. It has four categories:

- Must-Have: Essential for the first release. The system cannot do its main job without it.
- Should-Have: Important, but the first release can still work if it is delayed.
- Could-Have: A useful extra feature, added only if time and resources remain.
- Won't-Have (this release): Not included in this release because of scope, cost or resources. It may be added later.

Procedure

Step 1: Create the workbook

Open Microsoft Excel and create a new workbook. Save it as Library_Management_System_MoSCoW.xlsx.

Step 2: Create the prioritization sheet

Rename the first sheet "MoSCoW Prioritization". Add the columns No., Requirement, MoSCoW Category, Impact on Users and Justification.

Step 3: Enter the requirements

Type the ten requirements given in the question, one in each row.

Step 4: Assign the categories

For each requirement, consider its impact on users and the time, budget and effort needed. Select Must-Have, Should-Have, Could-Have or Won't-Have and write a short justification.

Step 5: Colour the categories

Use a different fill colour for each category (green, yellow, blue and orange) so that the categories are easy to identify.

Step 6: Create the guide and summary sheets

Add a sheet "MoSCoW Guide" explaining each category. Add a sheet "Summary" and use the COUNTIF formula to count the requirements in each category.

Step 7: Save the workbook

Review the entries and save the workbook.

Final Categorization

No.	Requirement	Category	Reason
1	Search books by title and author	Must-Have	Core function; easy to build with a database search.
2	Online book reservation system	Must-Have	Lets users reserve books; a basic version is feasible.
3	Monthly reports on borrowed books	Should-Have	Helpful, but can be built later from records.
4	Email notifications for overdue books	Should-Have	Reduces late returns; needs email setup.
5	QR code scanning for borrowing and returning	Could-Have	Speeds up borrowing; needs camera support and testing.
6	User-friendly dashboard for librarians	Must-Have	Needed for daily work; a basic one is easy to build.
7	User reviews and ratings	Could-Have	Not needed for borrowing; needs moderation.
8	Chatbot for user assistance	Could-Have	Needs setup and maintenance; an FAQ page can replace it.
9	Mobile app version	Won't-Have (this release)	Needs more time and money; a responsive website can serve for now.
10	Multi-language support	Should-Have	Improves accessibility; can be added in a later release.

Note: The categories are based on a student project with limited time, budget and team size. They may change if the scope or resources change.

Excel Worksheets

The following figures show the three sheets of the completed Excel workbook.

	A	B	C	D	E
1	**No.**	**Requirement**	**MoSCoW Category**	**Impact on Users**	**Justification (Impact and Feasibility)**
2	1	Search books by title and author	Must-Have	High	Core library function; users cannot find books without it. Easy to build with a database search.
3	2	Online book reservation system	Must-Have	High	Lets users reserve books without visiting the library. A basic version is feasible.
4	3	Monthly reports on borrowed books	Should-Have	High for administrators	Helps planning, but the system can work without it at first. Reports can be built from records later.
5	4	Email notifications for overdue books	Should-Have	Medium-high	Reduces late returns, but needs email setup. It can follow the core features.
6	5	QR code scanning for borrowing and returning	Could-Have	Medium-high	Speeds up borrowing but needs camera support and extra testing. Manual entry works for now.
7	6	User-friendly dashboard for librarians	Must-Have	High	Needed for daily library work. A basic dashboard is easy to build.
8	7	User reviews and ratings	Could-Have	Medium	Nice to have but not needed for borrowing. It also needs moderation.
9	8	Chatbot for user assistance	Could-Have	Medium	Helpful, but needs setup and maintenance. An FAQ page can replace it at first.
10	9	Mobile app version	Won't-Have (this release)	Potentially high	A separate app needs more time and money. A responsive website can serve users for now.
11	10	Multi-language support	Should-Have	Medium-high	Improves accessibility, but translation and testing can be added in a later release.

Figure 1: MoSCoW Prioritization sheet

	A	B
1	**Category**	**Meaning**
2	Must-Have	Essential for the first release. The system cannot do its main job without it.
3	Should-Have	Important, but the first release can still work if it is delayed.
4	Could-Have	Useful extra feature, added only if time and resources remain.
5	Won't-Have (this release)	Not included in this release because of scope, cost or resources. It may be added later.

Figure 2: MoSCoW Guide sheet

	A	B
1	**Category**	**Number of Requirements**
2	Must-Have	3
3	Should-Have	3
4	Could-Have	3
5	Won't-Have (this release)	1
6	**Total**	**10**

Figure 3: Summary sheet

Result

The ten Library Management System requirements were categorized using the MoSCoW method in Excel. Three requirements are Must-Have, three are Should-Have, three are Could-Have and one is Won't-Have for this release. Each category was justified based on user impact and feasibility.